MaxSup

Overcoming Representation Collapse in Label Smoothing

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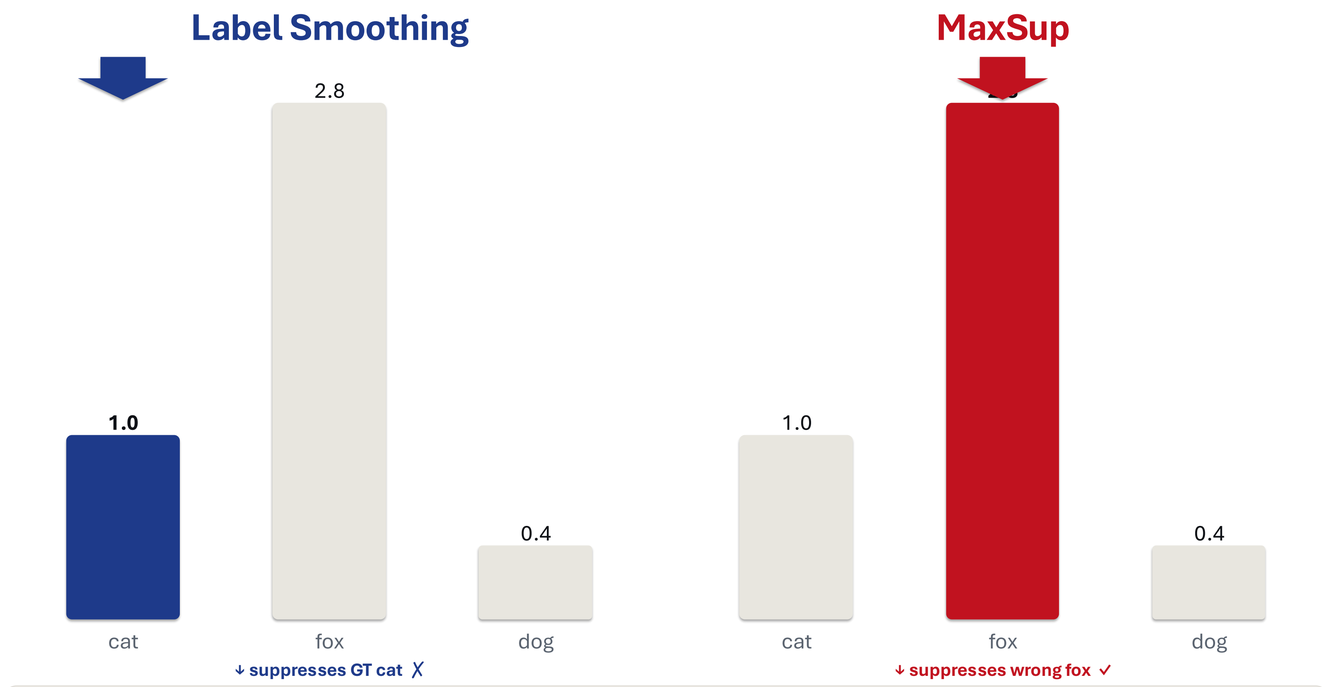
COMP7404 · HKU

"The class that should be regularized is not always the labeled class — it's the class the model currently trusts most."

THE CORE INSIGHT

How each method responds




```
L_LS = α ( z_gt − mean(z) )
L_MaxSup = α ( z_max − mean(z) )  ← one symbol changed
```

LS penalizes the labeled class even when the model is already wrong. MaxSup penalizes whichever class the model currently trusts most — a consistent signal in both correct and wrong predictions.

PROBLEM & METHOD

THE PROBLEM

LS decomposes into two terms:

Regularization term

Acts on logits below z_gt.
Reduces overconfidence. Good.

Error amplification term

Acts on logits above z_gt.
When model is wrong, widens the gap. Bad.

On misclassified samples, the bad term dominates.

THE FIX

One substitution eliminates the bad term:

z_gt → z_max

replace ground-truth logit with top-1 logit

Correct prediction

z_max = z_gt
Acts exactly like confidence regularization.

Wrong prediction

z_max ≠ z_gt
Directly suppresses the wrong dominant class.

Consistent regularization signal regardless of correctness.

RESULTS

DeiT-Small ImageNet

76.49%

+0.41 over LS
+2.10 over Baseline
Works on Transformers, not just CNNs.

ADE20K Segmentation

42.8 mIoU

+0.4 over LS
Backbone features transfer better to dense prediction tasks.

Grad-CAM Attention

More focused

LS attends to background cues.
MaxSup attends to the object body.
Better visual interpretability.

WHY IT MATTERS

[1] Plug-in

No architecture change. Drop-in loss replacement.

[2] Consistent

Same correct signal on all samples, right or wrong.

[3] Transferable

Better features for segmentation, fine-grained, and OOD tasks.

[4] Interpretable

Model attends to the right regions (Grad-CAM evidence).

MaxSup: Overcoming Representation Collapse in Label Smoothing | arXiv 2502.15798

Limitation: low-shot class performance remains an open challenge.

Demo
QR

Simple.
Focused.
Better.